

44.1 The additive model

Suppose a response depends on five continuous covariates and nothing is known about the shape of that dependence. The honest model is $E(Y | \mathbf{z}) = f(z_1, \dots, z_5)$, f unknown and smooth. Chapter 43 estimated such an f from one covariate by averaging the responses near the point of interest; in five dimensions that fails, for a reason that has nothing to do with the estimator.

44.1.1 Why additivity

Take n points spread over the unit cube in q dimensions. To collect a fixed fraction c of them, a cube-shaped neighbourhood must have side $c^{1/q}$: with $c = 0.05$ that is 0.05 in one dimension, about 0.55 in five and about 0.74 in ten. An average over three quarters of the range of every covariate is not local, and no estimator repairs that: the data are not dense enough. This is the *curse of dimensionality*, and the rates make it quantitative. For a twice-differentiable curve Theorem 43.1.3 gives a mean squared error of order $n^{-4/5}$; in q dimensions the same expansion gives $n^{-4/(4+q)}$, and no estimator does better for twice-differentiable regression functions (Stone, 1980, whose conditions and proof are not reproduced here). At $q = 5$ a twelve-fold increase in sample size buys only what a four-fold increase buys in one dimension.

The response is to restrict the shape of f — not to a linear function, usually too severe a restriction, but to a sum of functions of one variable each.

Definition 44.1.1. Additive model

Let $(y_i, \mathbf{x}_{(i)}, z_{i1}, \dots, z_{iq})$, $i = 1, \dots, n$, be observed, where $\mathbf{x}_{(i)}^\top$ is the i th row of an $n \times p$ matrix $\mathbf{X}$ whose first column is $\mathbf{1}$. The **additive model** is

$$y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \sum_{j=1}^q f_j(z_{ij}) + \varepsilon_i, \quad E(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad (44.1.1)$$

with the ε_i uncorrelated, the f_j unknown smooth functions, and the **identifiability constraints**

$$\sum_{i=1}^n f_j(z_{ij}) = 0, \quad j = 1, \dots, q. \quad (44.1.2)$$

The vector $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} + \sum_j \mathbf{f}_j$, where $\mathbf{f}_j = \{f_j(z_{1j}), \dots, f_j(z_{nj})\}^\top$, is the **additive predictor**. If $\mathbf{X} = \mathbf{1}$ the model is *purely additive*; otherwise it is a **partial linear** or semiparametric model.

Two comments at once. The model contains no interactions: the effect of z_1 is the same curve whatever z_2 does, and Section 44.3 buys some of that flexibility back. And Eq. 44.1.2 is not a restriction on the model but a choice of representation, because the additive predictor fixes the pieces only up to constants.

Proposition 44.1.1. Identifiability of the components

Write $\mathcal{V}_j$ for the linear space of vectors $\{g(z_{1j}), \dots, g(z_{nj})\}^\top$ as g ranges over the functions the j th term is allowed to take, and assume $\mathbf{1} \in \mathcal{V}_j$ for each j .

- (a) If $(\boldsymbol{\beta}, f_1, \dots, f_q)$ and $(\tilde{\boldsymbol{\beta}}, \tilde{f}_1, \dots, \tilde{f}_q)$ give the same additive predictor and both satisfy Eq. 44.1.2, then every difference $\mathbf{f}_j - \tilde{\mathbf{f}}_j$ is orthogonal to $\mathbf{1}$, and so is $\mathbf{X}(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}) = -\sum_j (\mathbf{f}_j - \tilde{\mathbf{f}}_j)$. If in addition the centred spaces $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, $\mathcal{V}_1 \cap \mathbf{1}^\perp, \dots, \mathcal{V}_q \cap \mathbf{1}^\perp$ admit no nontrivial relation — no *exact concurrency*, the condition studied in Section 44.2 — then $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\tilde{\boldsymbol{\beta}}$ and $\mathbf{f}_j = \tilde{\mathbf{f}}_j$ for every j .
- (b) Without Eq. 44.1.2 the components are determined only up to constants that sum to zero: replacing f_j by $f_j + c_j$ with $\sum_j c_j = 0$ leaves $\boldsymbol{\eta}$ unchanged.
- (c) If $\mathbf{f}_j = \mathbf{Z}_j \boldsymbol{\gamma}_j$ for a design block $\mathbf{Z}_j$, the constraint $\mathbf{1}^\top \mathbf{Z}_j \boldsymbol{\gamma}_j = 0$ is imposed exactly by writing $\boldsymbol{\gamma}_j = \mathbf{U}_j \boldsymbol{\alpha}_j$, where the columns of $\mathbf{U}_j$ are an orthonormal basis of $\{\boldsymbol{\gamma} : \mathbf{1}^\top \mathbf{Z}_j \boldsymbol{\gamma} = 0\}$: minimizing any criterion over $\boldsymbol{\alpha}_j$ with design $\mathbf{Z}_j \mathbf{U}_j$ and penalty $\mathbf{U}_j^\top \mathbf{P}_j \mathbf{U}_j$ is the same as minimizing it over $\boldsymbol{\gamma}_j$ subject to the constraint.

Proof.

- (b) is immediate from $\sum_j (f_j + c_j) = \sum_j f_j + \sum_j c_j$. For (a), equality of the predictors gives $\mathbf{X}(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}) = -\sum_j (\mathbf{f}_j - \tilde{\mathbf{f}}_j)$. Each difference $\mathbf{f}_j - \tilde{\mathbf{f}}_j$ lies in $\mathcal{V}_j \cap \mathbf{1}^\perp$, because both terms satisfy Eq. 44.1.2, so the left-hand side lies in $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ and the displayed equation is a relation among the centred spaces. That alone does not separate the pieces. Under the additional hypothesis the relation must be trivial, so $\mathbf{X}(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}) = \mathbf{0}$ and every difference vanishes.

For (c), $\{\boldsymbol{\gamma} : \mathbf{1}^\top \mathbf{Z}_j \boldsymbol{\gamma} = 0\}$ is the null space of the row vector $\mathbf{c}^\top = \mathbf{1}^\top \mathbf{Z}_j$, of dimension $d_j - 1$ when $\mathbf{c} \neq \mathbf{0}$, and $\mathbf{U}_j$ is a bijection from $\mathbb{R}^{d_j-1}$ onto it. At $\boldsymbol{\gamma}_j = \mathbf{U}_j \boldsymbol{\alpha}_j$ the design term is $(\mathbf{Z}_j \mathbf{U}_j) \boldsymbol{\alpha}_j$ and the penalty $\boldsymbol{\alpha}_j^\top (\mathbf{U}_j^\top \mathbf{P}_j \mathbf{U}_j) \boldsymbol{\alpha}_j$, so the two minimizations agree at corresponding points and the correspondence is onto the feasible set. ■

A QR factorization of $\mathbf{Z}_j^\top \mathbf{1}$ produces $\mathbf{U}_j$ in one line: the first column of $\mathbf{Q}$ spans $\mathbf{c}$, the rest its orthogonal complement. Absorbing the constraint this way rather than centring the columns of $\mathbf{Z}_j$ matters, because a basis with the partition-of-unity property of Proposition 43.2.1 has $\mathbf{Z}_j \mathbf{1} = \mathbf{1}$: subtracting column means leaves $\mathbf{Z}_j$ with a null vector and the penalized cross-product singular (Exercise (B1)).

44.1.2 Terms, designs and penalties

Every term in this chapter is represented in a basis. Choose d_j basis functions for the j th component and write

$$f_j(z) = \sum_{l=1}^{d_j} \gamma_{jl} B_{jl}(z), \quad \mathbf{f}_j = \mathbf{Z}_j \boldsymbol{\gamma}_j, \quad (\mathbf{Z}_j)_{il} = B_{jl}(z_{ij}), \quad (44.1.3)$$

after which the constraint of Proposition 44.1.1(c) has been absorbed, so that $\mathbf{Z}_j$ has d_j columns summing to zero. With the B-spline basis of Definition 43.2.1 and the difference penalty of Definition 43.3.1 this is a P-spline block, the default term throughout the chapter; but nothing below uses that particular basis. All that is used is that the roughness of the j th component is a quadratic form $\boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j$ with $\mathbf{P}_j$ symmetric nonnegative definite.

The estimate is then the minimizer of a penalized least squares criterion in the sense of Defini-

tion 30.1.1, with one penalty per term:

$$Q(\boldsymbol{\beta}, \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q) = \left\| \mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \sum_{j=1}^q \mathbf{Z}_j \boldsymbol{\gamma}_j \right\|^2 + \sum_{j=1}^q \lambda_j \boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j. \quad (44.1.4)$$

Collect the pieces: let $\mathbf{Z} = [\mathbf{X}, \mathbf{Z}_1, \dots, \mathbf{Z}_q]$ with $d = p + \sum_j d_j$ columns, let $\boldsymbol{\gamma} = (\boldsymbol{\beta}^\top, \boldsymbol{\gamma}_1^\top, \dots, \boldsymbol{\gamma}_q^\top)^\top$, and let

$$\mathbf{P} = \begin{pmatrix} \mathbf{0} & & & \\ & \lambda_1 \mathbf{P}_1 & & \\ & & \ddots & \\ & & & \lambda_q \mathbf{P}_q \end{pmatrix} \quad (44.1.5)$$

be the block-diagonal penalty, with a zero block for the unpenalized columns. Then Eq. 44.1.4 is $\|\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}\|^2 + \boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma}$, a ridge-type criterion with a structured penalty, and differentiating gives the **penalized normal equations**

$$\mathbf{A}\boldsymbol{\gamma} = \mathbf{Z}^\top \mathbf{y}, \quad \mathbf{A} = \mathbf{Z}^\top \mathbf{Z} + \mathbf{P}. \quad (44.1.6)$$

Example. Two P-spline blocks

Take $q = 2$, a cubic B-spline basis of 15 functions on equally spaced knots for each covariate, and the second-difference penalty of Definition 43.3.1. Unconstrained, each block has 15 columns summing to $\mathbf{1}$ row by row and a penalty of rank 13; after Proposition 44.1.1(c) it has 14 columns summing to zero and a penalty of the same rank, so $\mathbf{Z} = [\mathbf{1}, \mathbf{Z}_1, \mathbf{Z}_2]$ has 29 columns. Of the two unpenalized directions per block, constant and linear trend, the constraint removes one: the term still fits a straight line in z_j free of charge, but not an arbitrary level.

44.1.3 The fit is a linear smoother

Because Eq. 44.1.4 is quadratic, the fit is linear in $\mathbf{y}$ — a linear smoother in the sense of Proposition 43.5.1 — and every quantity of interest is a trace, exactly as for ridge regression.

Proposition 44.1.2. Smoother matrix and degrees of freedom

Suppose $\mathbf{A} = \mathbf{Z}^\top \mathbf{Z} + \mathbf{P}$ is nonsingular, and let $\mathbf{C} = \mathbf{A}^{-1} \mathbf{Z}^\top$, so that $\hat{\mathbf{y}} = \mathbf{C}\mathbf{y}$. Write $\mathbf{C}_j = \mathbf{Z}_j \mathbf{C}_{[j]}$, where $\mathbf{C}_{[j]}$ holds the rows of $\mathbf{C}$ belonging to the j th block.

- (a) The fitted vector is $\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$ with $\mathbf{S} = \mathbf{Z}\mathbf{A}^{-1}\mathbf{Z}^\top$, a symmetric matrix whose eigenvalues lie in $[0, 1]$; the fitted j th component is $\hat{\mathbf{f}}_j = \mathbf{C}_j \mathbf{y}$, and $\mathbf{S} = \mathbf{X}\mathbf{C}_{[0]} + \sum_j \mathbf{C}_j$.
- (b) The **effective degrees of freedom** of the fit and of its terms are

$$\text{df} = \text{tr } \mathbf{S} = \text{tr}(\mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Z}), \quad \text{df}_j = \text{tr } \mathbf{C}_j = d_j - \lambda_j \text{tr}\{(\mathbf{A}^{-1})_{jj} \mathbf{P}_j\}, \quad (44.1.7)$$

and they add: $\text{df} = p + \sum_j \text{df}_j$. Each df_j decreases from d_j at $\lambda_j = 0$ towards $\dim \mathcal{N}(\mathbf{P}_j)$ as $\lambda_j \rightarrow \infty$.

(c) If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $\text{Cov}(\hat{\mathbf{y}}) = \sigma^2 \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Z} \mathbf{A}^{-1}$ and $\text{Cov}(\hat{\mathbf{f}}_j) = \sigma^2 \mathbf{C}_j \mathbf{C}_j^\top$.

Proof.

- (a) Solving Eq. 44.1.6 gives $\hat{\mathbf{y}} = \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{y}$ and $\hat{\mathbf{y}} = \mathbf{Z} \hat{\mathbf{y}} = \mathbf{S} \mathbf{y}$. Symmetry is clear, and $\mathbf{v}^\top \mathbf{S} \mathbf{v} \geq 0$ because $\mathbf{A}^{-1}$ is positive definite. For the upper bound, fix $\mathbf{v}$ and put $\mathbf{u} = \mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{v}$, so that $\mathbf{A} \mathbf{u} = \mathbf{Z}^\top \mathbf{v}$. Then

$$\mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{u}^\top \mathbf{A} \mathbf{u} = \|\mathbf{Z} \mathbf{u}\|^2 + \mathbf{u}^\top \mathbf{P} \mathbf{u} \geq \|\mathbf{Z} \mathbf{u}\|^2,$$

while also $\mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{v}^\top (\mathbf{Z} \mathbf{u}) \leq \|\mathbf{v}\| \|\mathbf{Z} \mathbf{u}\| \leq \|\mathbf{v}\| \sqrt{\mathbf{v}^\top \mathbf{S} \mathbf{v}}$ by Cauchy–Schwarz. Dividing by $\sqrt{\mathbf{v}^\top \mathbf{S} \mathbf{v}}$ when it is nonzero gives $\mathbf{v}^\top \mathbf{S} \mathbf{v} \leq \|\mathbf{v}\|^2$, so the eigenvalues lie in $[0, 1]$. Splitting $\mathbf{Z} \hat{\mathbf{y}}$ by blocks gives the last identity.

- (b) $\text{tr} \mathbf{S} = \text{tr}(\mathbf{Z} \mathbf{A}^{-1} \mathbf{Z}^\top) = \text{tr}(\mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Z})$ by the cyclic property (Theorem 1.6.1), and $\mathbf{A}^{-1} \mathbf{Z}^\top \mathbf{Z} = \mathbf{A}^{-1}(\mathbf{A} - \mathbf{P}) = \mathbf{I} - \mathbf{A}^{-1} \mathbf{P}$. The j th diagonal block of $\mathbf{A}^{-1} \mathbf{P}$ is $\lambda_j(\mathbf{A}^{-1})_{jj} \mathbf{P}_j$, and the block belonging to the unpenalized columns is zero; taking traces block by block gives both formulas in Eq. 44.1.7 and their sum. Monotonicity in λ_j is Exercise (B2), and as $\lambda_j \rightarrow \infty$ the term is forced into $\mathcal{N}(\mathbf{P}_j)$, which costs $\dim \mathcal{N}(\mathbf{P}_j)$ dimensions.
- (c) Both are immediate from $\text{Cov}(\mathbf{M} \mathbf{Y}) = \mathbf{M} \text{Cov}(\mathbf{Y}) \mathbf{M}^\top$ (Theorem 2.2.1). ■

Part (b) is the additive-model version of Theorem 27.2.2(b): each term is charged for the dimensions it uses, and the charges add up. The split makes a printed summary readable — “living area: 5.3 degrees of freedom” says at a glance how far from a straight line the curve is — and it is what Chapter 29 needs, since Definition 29.3.2 and Proposition 29.2.4 apply verbatim with df in place of the number of parameters.

Warning. The degrees of freedom of a *term* are not the degrees of freedom of a test: they are a trace computed at a fixed $\boldsymbol{\lambda}$, and nothing so far says that df_j is the right divisor for a sum of squares or the right parameter for a reference distribution. Section 44.6 returns to it.

44.1.4 Exercises

44.1.4.1 A. Check your understanding

Exercise. A1

Repeat the neighbourhood calculation of Section 44.1: what side length does a cube-shaped neighbourhood need in order to contain 1% of n points uniformly spread over the unit cube in q dimensions, for $q = 1, 3, 10$? At what q does the required side exceed 0.5?

44.1.4.2 B. Practice

Exercise. B1

Let $\mathbf{Z}_j$ be a B-spline block with the partition-of-unity property, so $\mathbf{Z}_j \mathbf{1} = \mathbf{1}$, and let $\mathbf{P}_j$ be a difference penalty, so $\mathbf{P}_j \mathbf{1} = \mathbf{0}$. Show that centring the *columns*, that is replacing $\mathbf{Z}_j$

by $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Z}_j$, makes $\mathbf{Z}_j^\top \mathbf{Z}_j + \lambda_j \mathbf{P}_j$ singular, while the reparameterization of [Proposition 44.1.1\(c\)](#) does not.

Solution. Write $\tilde{\mathbf{Z}}_j = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Z}_j$. Then $\tilde{\mathbf{Z}}_j \mathbf{1} = \mathbf{0}$, so $\mathbf{1}^\top (\tilde{\mathbf{Z}}_j^\top \tilde{\mathbf{Z}}_j + \lambda_j \mathbf{P}_j) \mathbf{1} = 0$ and the matrix is singular. Under (c) the matrix is $\mathbf{U}_j^\top (\mathbf{Z}_j^\top \mathbf{Z}_j + \lambda_j \mathbf{P}_j) \mathbf{U}_j$; if it killed some $\boldsymbol{\alpha} \neq \mathbf{0}$, then $\boldsymbol{\gamma} = \mathbf{U}_j \boldsymbol{\alpha}$ would satisfy $\mathbf{Z}_j \boldsymbol{\gamma} = \mathbf{0}$ and $\mathbf{P}_j \boldsymbol{\gamma} = \mathbf{0}$. The second puts $\boldsymbol{\gamma}$ in the span of the constant, or of the constant and linear sequences for a second-difference penalty, and then $\mathbf{Z}_j \boldsymbol{\gamma} = \mathbf{0}$ forces $\boldsymbol{\gamma} = \mathbf{0}$, since the B-spline basis reproduces those functions exactly.

Exercise. B2

Show that df_j in [Eq. 44.1.7](#) is nonincreasing in λ_j . *Hint:* work with one term and no parametric part first, where $\text{df}(\lambda) = \sum_k \theta_k / (\theta_k + \lambda)$ for the eigenvalues θ_k of $\mathbf{Z}_j^\top \mathbf{Z}_j$ in the metric of $\mathbf{P}_j$.

Solution. Simultaneously diagonalize $\mathbf{Z}_j^\top \mathbf{Z}_j$ and $\mathbf{P}_j$: some nonsingular $\mathbf{T}$ gives $\mathbf{T}^\top \mathbf{Z}_j^\top \mathbf{Z}_j \mathbf{T} = \mathbf{I}$ and $\mathbf{T}^\top \mathbf{P}_j \mathbf{T} = \text{diag}(\nu_k)$, $\nu_k \geq 0$, by Corollary 1.7.8(b). Then $\text{df}(\lambda) = \sum_k 1/(1 + \lambda \nu_k)$, decreasing in λ except where $\nu_k = 0$, which contributes $\dim \mathcal{N}(\mathbf{P}_j)$ whatever λ is. With several terms, argue the same way on the j th block of $\mathbf{A}^{-1} \mathbf{P}$, treating the others as part of $\mathbf{Z}^\top \mathbf{Z}$.

Exercise. B3

In [Definition 44.1.1](#) take $q = 1$ and let $\mathbf{X}$ be a general parametric design. Show that the penalized estimate of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = \{\mathbf{X}^\top (\mathbf{I} - \mathbf{S}_1) \mathbf{X}\}^{-1} \mathbf{X}^\top (\mathbf{I} - \mathbf{S}_1) \mathbf{y}$ where $\mathbf{S}_1 = \mathbf{Z}_1 (\mathbf{Z}_1^\top \mathbf{Z}_1 + \lambda_1 \mathbf{P}_1)^{-1} \mathbf{Z}_1^\top$ — the Frisch–Waugh–Lovell form of Theorem 6.6.2 with the smoother in place of a projection. What changes in the proof?

44.1.4.3 C. Going deeper

Exercise. C1

An additive model has no interactions. Show that if the true mean is $f(z_1, z_2) = z_1 z_2$ on $[0, 1]^2$ with z_1, z_2 independent and uniform, the best additive approximation in mean square, $\arg \min_{g_1, g_2} \mathbb{E}\{(Z_1 Z_2 - g_1(Z_1) - g_2(Z_2))^2\}$ subject to $\mathbb{E} g_1(Z_1) = \mathbb{E} g_2(Z_2) = 0$, is $g_1(z) = (z - \frac{1}{2})/2$, $g_2(z) = (z - \frac{1}{2})/2$ plus the constant $1/4$, and compute the mean squared approximation error.